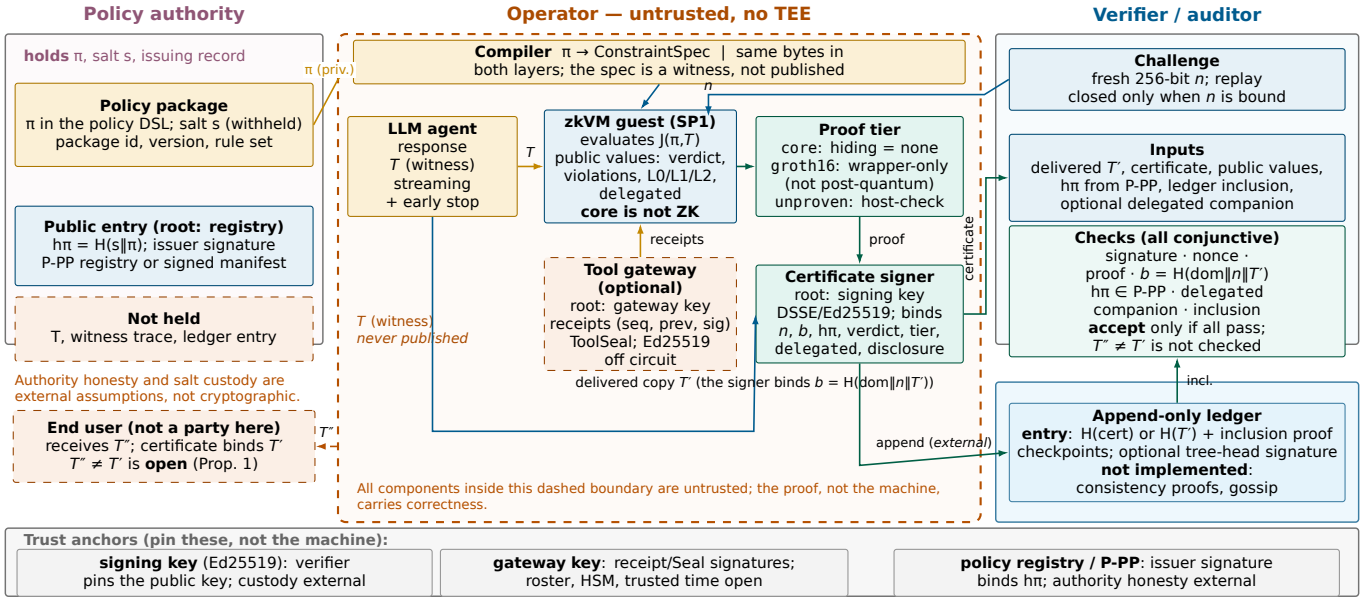
Policy authority
Operator — untrusted, no TEE
Verifier / auditor
holds π, salt s, issuing record
π (priv.)
Policy package
π in the policy DSL; salt s (withheld)
package id, version, rule set
Public entry (root: registry)
hπ = H(s‖π); issuer signature
P-PP registry or signed manifest
Not held
T, witness trace, ledger entry
Authority honesty and salt custody are external assumptions, not cryptographic.
End user (not a party here)
receives T″; certificate binds T′
T″ ≠ T′ is open (Prop. 1)
T″
Compiler π → ConstraintSpec | same bytes in both layers; the spec is a witness, not published
n
LLM agent
response
T (witness)
streaming
+ early stop
T
zkVM guest (SP1)
evaluates J(π,T)
public values: verdict, violations, L0/L1/L2, delegated
core is not ZK
Proof tier
core: hiding = none
groth16: wrapper-only (not post-quantum)
unproven: host-check
receipts
proof
Tool gateway (optional)
root: gateway key
receipts (seq, prev, sig)
ToolSeal; Ed25519
off circuit
Certificate signer
root: signing key
DSSE/Ed25519; binds n, b, hπ, verdict, tier, delegated, disclosure
certificate
T (witness)
never published
delivered copy T′ (the signer binds b = H(dom‖n‖T′))
append (external)
All components inside this dashed boundary are untrusted; the proof, not the machine, carries correctness.
Challenge
fresh 256-bit n; replay
closed only when n is bound
Inputs
delivered T′, certificate, public values, hπ from P-PP, ledger inclusion, optional delegated companion
Checks (all conjunctive)
signature · nonce ·
proof · b = H(dom‖n‖T′)
hπ ∈ P-PP · delegated companion · inclusion
accept only if all pass;
T″ ≠ T′ is not checked
incl.
Append-only ledger
entry: H(cert) or H(T′) + inclusion proof
checkpoints; optional tree-head signature
not implemented:
consistency proofs, gossip
Trust anchors (pin these, not the machine):
signing key (Ed25519): verifier pins the public key; custody external
gateway key: receipt/Seal signatures; roster, HSM, trusted time open
policy registry / P-PP: issuer signature binds hπ; authority honesty external
public
witness
signed / anchored
external / open
crypto binding
external assumption
open gap
Design, not measured. No TEE. core leaves the witness open; groth16 hiding is wrapper-only and not post-quantum; L-side (re-execution) is open at every tier.
External: authority holds s; operator does not leak π/T; gateway key custody; append-before-delivery and no split view; T″ binding to the end user.